Stack Demo
Test Plan

E-Commerce Web Application Testing

https://bstackdemo.com/

1. Target Platform URL

https://bstackdemo.com/

2. System Description

Stack Demo is an e-commerce web platform used to test core functionalities such as user authentication, product catalog browsing, vendor filtering, cart management, checkout workflow, and order history tracking.

3. Testing Focus

Verifying core functional modules and ensuring system stability across manual, automation, API, and cross-browser test execution cycles.

4. Project Purpose — Core Purpose & Quality Objectives

Item	Core Purpose & Quality Objectives
Objective 1	Verify that all user-level shopping workflows work as expected.
Objective 2	Ensure data integrity across UI, API, and cart calculation layers.
Objective 3	Identify UI, functional, cross-browser, and API defects.
Objective 4	Validate checkout simulation, shipping details entry, and order confirmation flows.

5. Blueprint vs Test Plan Element

Blueprint Element	Test Plan Element
Building Design	Test Objectives: Ensure all major e-commerce features (login, product filtering, add to cart, checkout) function correctly. Verify the website meets both functional and non-functional requirements. Identify, log, and track defects using Jira. Validate stability, usability, and cross-browser compatibility before release.

6. Scope of Testing

Scope Category	Detailed Functionality Coverage
In Scope	• User registration & login (Standard, Locked, Existing orders, Fav users). • Product search & vendor filtering (Apple, Samsung, Google, OnePlus). • Price sorting (Lowest to Highest, Highest to Lowest). • Add to Cart & Wishlist / Favourites. • Checkout process & shipping form validation. • Order confirmation overlay & order history management. • API endpoint testing (products, product detail, signin, cart). • Cross-browser testing (Chrome, Firefox, Edge). • Cart persistence across page refresh. • Sort + filter combination behaviour. • Security testing (XSS, SQL Injection). • Edge Cases & Form Validations (ZIP code, blank spaces, max quantity).
Interfaces & Workflows	• UI testing through web interface across supported desktop browsers. • API interface through Python requests library. • Workflow: Product Catalog → Filter/Sort → Add to Cart → Checkout → Confirmation. • Vendor Integration: Mock product feeds (Stack Demo internal data).
Out of Scope	• Real payment gateway transaction clearance with real banking institutions. • Native mobile application testing (iOS / Android native binaries). • High-volume performance and load stress testing. • Unit testing / Database (MySQL) validation. • Admin dashboard testing.

7. Testing Strategy & Types

Type	Description
Manual testing	Functional, Exploratory, and Documented End-to-End User Scenario Testing.
Automation testing	Automate regression test cases for login, product filtering, cart, and checkout modules using Selenium WebDriver (Python).
API testing	Validate REST endpoints (products, signin, cart) using Python requests.
Cross-browser testing	Run the full suite on Chrome, Firefox, and Edge.
Non-functional testing	Basic usability and cross-browser compatibility checks. High-volume load testing remains out of scope.
Security testing	Basic XSS and SQL Injection payload checks on forms and APIs.

8. Testing Strategy — Responsibilities Example

Role	Responsibilities
Test Team	Execute test cases, report defects.
Technical Team	Resolve defects and deploy builds.
Project Manager	Monitor progress and approve releases.

9. Test Assumptions & Data Approach

Parameter	Details & Guidelines
Test Assumptions	• Test environment is stable and accessible at https://bstackdemo.com/. • All required modules are functional on the demo platform. • Dummy user credentials and sample product data are available. • Internet connection and target browser versions are available. • Test cases must be validated against actual UI behaviour before being marked as FAIL.
Data Approach	• Test data will be created manually and maintained in the QA repository. • Functional Testing: Dummy users, test products, and fake shipping addresses. • UAT Testing: Realistic user scenarios with valid and invalid inputs. • No confidential or production data will be used.

10. Test Environment Specifications

Parameter	Value
URL	https://bstackdemo.com/
Browser	Google Chrome, Microsoft Edge, Mozilla Firefox
Automation tool	Selenium WebDriver (Python)
Backend	RESTful Web Services / Session Storage
OS	Windows 10–11, macOS

11. Level of Testing — Functional & UAT

Testing Level	Features to be Tested
Functional Testing	• User authentication (standard login, locked account error checks, logout). • Product catalog filtering, brand multi-select, price sort order. • Adding, updating item counts, and removing items in shopping cart. • Checkout process: shipping address input, submit action, modal popup. • Order confirmation display and order history tracking. • Cart persistence across page refresh. • Sort + filter combination. • Security forms validation (XSS, SQL Injection).

Testing Level	Features to be Tested
User Acceptance Testing (UAT)	• User Account: Validate user profile selection, login, logout, and order history view. • Product & Cart: Validate filtering, adding items, modifying quantities, and removing items. • Checkout: Validate address entry, place order action, and confirmation display. • Order Management: Confirm placed order appears correctly under existing_orders_user.

12. Level of Testing — Regression Testing Matrix

Focus Area	Regression Validation Rules
Authentication	Verify user authentication and login still function correctly after system updates or code commits.
Cart & Favourites	Validate that adding/removing items in the cart and favourites continues to update totals accurately.
Price Calculations	Ensure quantity updates and price recalculations remain completely accurate.
Data & Layout	Ensure product images, prices, vendor tags, and stock information display consistently.
Cross-Browser UI	Validate data consistency between UI display and API responses across Chrome, Firefox, and Edge.

13. Execution Strategy (Entry Criteria Guidelines)

Prerequisite Phase	Entry Benchmark Conditions
Environment Readiness	Test environment (https://bstackdemo.com/) is fully accessible, responsive, and stable.
Test Data Preparation	Test data for users, products, quantities, and addresses are created, verified, and ready.
Toolchain Setup	All testing tools (Selenium, requests, Jira, GitHub) are installed, configured, and verified.
Build Stability	The latest build is deployed successfully, and core modules (Login, Cart, Checkout) pass smoke checks.
If Conditions Are Not Met	The QA team will assess the associated risks, identify mitigation actions, and recommend whether to proceed or delay testing.

Entry Criteria Sign-Off Checklist

Entry Criteria	Test Team
Test environment(s) is available and accessible	Jana Tamer Hassan
Test data is created and validated	John Samy
Code build has been merged and deployed successfully	Basem Sherif
Development team completed initial smoke/unit checks	Basem Sherif
Test scripts are completed, reviewed, and approved by QA Lead	Basem Sherif

14. Execution Strategy (Exit Criteria Guidelines)

Quality Gate	Mandatory Exit Conditions
Execution Coverage	All planned test cases have been executed across functional, regression, UAT, and API cycles.
Pass Rate Target	At least 90% of test cases have passed successfully.
Defect Threshold	No Critical or High severity defects remain open in the issue tracker. All Medium and Low defects are logged in Jira.

Quality Gate	Mandatory Exit Conditions
Documentation & Cleanup	All test results and execution logs are stored; final reports are signed off; environment cleaned up.
If Criteria Are Not Met	The QA team will assess the impact and release risk, then recommend mitigation actions or follow-up testing.

Exit Criteria Sign-Off Matrix

Exit Criteria Benchmark	Responsible Entity
100% Test Scripts executed across all planned cycles	Jana Tamer Hassan
90% pass rate achieved across all executed test scripts	Basem Sherif
Zero open Critical and High severity defects remaining	Basem Sherif
All remaining defects are either cancelled or documented as Change Requests for future releases	Basem Sherif
All expected and actual results are captured and documented with test scripts	Farah Hossam El-Din
All test metrics collected based on daily and weekly status reports	Basem Sherif
All defects logged in Jira / Defect Tracking Spreadsheet	Fawzy Mohamed
Test environment cleanup completed and backup created	Jana Tamer Hassan

15. Validation & Defect Management Protocol

Aspect	Details
Execution	It is expected that testers will execute all assigned scripts in each cycle described in this plan.
Tracking	All discovered defects will be tracked using Jira.
Responsibility	It is the responsibility of the tester to open the defect ticket, retest upon fix, and officially close the ticket.
Verification	QA Lead verifies critical fixes before closing.

16. Timeline & Milestones

Time Period	Milestone & Core Activities
Week 1	Test Planning & Setup: Requirements analysis, Test Plan creation, environment verification, test data preparation, and Jira sprint creation.
Week 2–3	Manual Testing Execution: Exploratory testing, manual test case execution (login, filtering, cart, checkout), and defect reporting in Jira.
Week 4	Automation & API Execution: Selenium WebDriver framework setup, script creation, regression execution, and API collection runs.
Week 5	System & UAT Support: Cross-browser testing (Chrome, Edge, Firefox), retesting bug fixes, regression suite execution, and end-to-end UAT validation.
Final Week (Week 6)	Test Closure: Final defect triage, test summary report generation, exit criteria sign-off, and artifact archiving.

17. Defect Severity Matrix

Severity	Impact & Stack Demo Scenarios
Critical	System crash or core functionality failure preventing complete usage. Examples: Checkout modal fails to open or crashes on submit; Add to Cart button completely unclickable; Login/Sign-in completely broken; Server error (500) appears on homepage.
High	Major feature not working properly, but system remains partially usable. Examples: Product images fail to render for specific vendor filters; Total price calculation fails to update on qty change; Order confirmation overlay displays wrong shipping address.
Medium	Minor functionality or layout issue that does not halt core purchase flows. Examples: Vendor filter checkbox misaligned; Product description overlaps border; Sorting dropdown slight loading lag.
Low	Cosmetic or low-priority issues with zero business or operational impact. Examples: Typo in product description; Slight color mismatch on footer copyright; Missing favicon on browser tab.

18. Project & Testing Dependencies

Dependency Type	Critical Requirement Details
Environment & Tools	Availability and stability of https://bstackdemo.com/; full access to Jira, Selenium, and Python requests.
Development & Builds	Stable builds and timely defect fixes from the development team before each test cycle.
Data & Scripts	Clean test data prepared prior to execution; functioning mock checkout simulation scripts.
Team Coordination	Timely defect fix confirmations; smooth coordination within the team for Git branching and PR reviews.

19. Risk Management & Mitigation

Category	Action Plan	Responsible
Identified Risks	Environment downtime, script locator breaking due to UI updates, delayed bug verification.	Basem Sherif
Mitigation Strategy	Maintain backup dynamic XPath/CSS, schedule daily team syncs, prioritize High/Critical defect triage.	Jana Tamer Hassan
Contingency Plan	Re-run failed automated cases manually in alternate browsers and maintain detailed DOM logs.	Fawzy Mohamed

20. Test Deliverables Matrix

#	Deliverable Artifact	Document Content & Scope
1	Test Plan Document	Test objectives, scope, strategy, in/out-of-scope definitions, testing types, schedule, team allocation.
2	Test Scenarios	High-level journeys: Registration, Login, Vendor Filtering, Cart, Checkout, Order History, API.
3	Test Cases Matrix (101)	Test Case ID, Title, Pre-conditions, Steps, Expected/Actual Results, Pass/Fail Status.
4	Test Data Set	Credentials (demouser, locked_user, existing_orders_user, fav_user), catalog params, addresses.
5	Test Execution Report	Total executed cases, pass/fail/blocked metrics, execution dates, tester names.
6	Bug / Defect Log (Jira)	Bug ID, summary, steps, expected/actual, severity/priority, status, reporter, assigned, environment.
7	Test Summary Report	Overview of testing process, defect distribution, pass rate metrics, release sign-off.
8	API Test Report	13 API test cases with endpoint, method, expected status, actual status.
9	Cross-Browser Report	Pass/fail matrix for Chrome, Firefox, Edge.
10	Automation Scripts	stackdemo_tests.py (Selenium + requests), requirements.txt.